PRIME MINISTERS' OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT MANYARA REGION

MANYARA REGIONAL MOCK

MANYARA REGION

041 **BASIC MATHEMATICS**

Time: 3Hours **AUGOST 2026**

Instructions

1. This paper consists of two sections **A** and **B** with a total of **fourteen (14)** questions

2. Answer **all questions** in sections A and B. Each question in section A carries **six (6) marks** while each question in section B carries **ten (10) marks**.

3. All necessary working and answers for each question must be shown clearly.

4. NECTA mathematical tables and non-programmable calculators may be used.

5. Cellular phones, calculators and any unauthorized materials are not allowed in the examination room.

6.The following constants may be used were necessary, the radius of the Earth R=6370KM and Π=3.14

7. Write your **Examination Number** on every page of your answer sheet.

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SECTION A: (60 MARKS)

ANSWER ALL QUESTIONS

1(a) Express $1.25\dot{}$ as a simple fraction in the form of a/b

(b) John used his salary as follows; 10% on house rent , 25% on food, 20% on refreshment and 15% on school fees. If he was left with Tshs. 150,000/= determine John's salary .

2(a) Without using mathematical tables simplify;

$\frac{\sqrt{63}-\sqrt{72}}{\sqrt{32}-\sqrt{28}}$ giving your answer in fraction form

b) Solve the equation $\log_2 x + \log_x 64 = 5$

3.(a) A box contains 6 red marbles and 4 blue marbles . If one marble is drawn from the box and put aside , then the second draw was done and replaced. What is the probability that

(i) They are of the same colour
(ii) One is blue and the other is red

(b) In Charles' class, 2pupils study English but not History , 4 study History but not English , I study neither History nor English . How many pupils study History ? (Given that there are only 20 pupils in Charles' class)

4. (a) A line perpendicular to y= 3 – 4x passes through the point (-8,5). Determine its equation.

(b) Find the magnitude of vector C if C = a + b and a = 2j, b=-I + 2j

5(a) The length of two side of a triangle are 18cm and 24 cm . Find the area of the triangle if the included angle is 30°

(b) The interior angle of a regular polygon is 60° greater than the exterior angle. Find the number of sides of the polygon.

6(a) A car uses18 litres of petrol for 154km. How many litres of petrol would it use for 231km?

(b) If x varies directly as Y inversely proportional as Z and X= 8, when Y=12 and Z= 6. Find the value of X when Y=16 and z=4.

7(a) Juma bought a car at 5,000,000/= and sold it for profit of 10%. Find

(i) The profit made
(ii) The selling price

(b) Prepare Trading Profit and loss Account of Mr Kassim enterprises Ltd as at 31/12/2003

MR KASSIM LTM TRIAL BALANCE

Account Name	**Dr**	**Cr**
Cash	1,900,000	
Capital		2,730,000
Purchase	2,200,000	
Rent	100,000	
Furniture	300,000	
Shelves	250,000	
Sales		2,130,000
Salary	60,000	
Wages	50,000	
	4,860,000	4,860,000

8.(a) Find the sum of the first 10 terms of the series .

$\text{Log}_3 x + \log_9 x + \log_{81} x + \ldots\ldots\ldots\ldots\ldots..$

(b) Find the amount of money accumulated at the end of 2 years after investing 500,000 shillings at a compound interest rate of 10% annually.

9 (a) A student stands at a distance of 10m from the foot of a tree. He measures the angle of elevation of the top of the tree and he finds that it is 45º. If his height is 1.5m tall, determine the height of the tree.

(b) A rectangular frame is made of wooden bars. The diagonal of the frame is 17cm long and its length is 15cm . Find the width of the frame .

10(.a) Solve the following quadratic equation by completing the square method

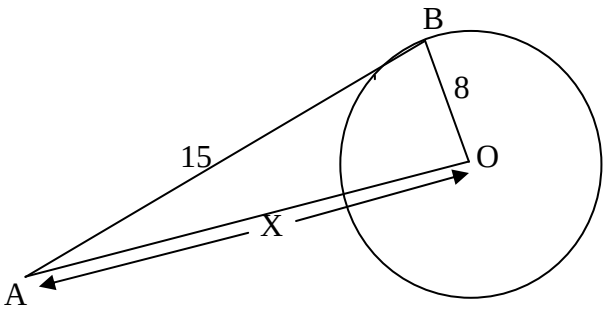
$2x^2 + x - 1 = 0$

(b) Factorise $a^2 - b^2$ and hence evaluate $2557^2 - 2547^2$

SECTION B (40 MARKS)

Answer **All** questions from this section

11. (a) In the figure given below find the value of x, given that O is the centre of the circle



(b) Carefully study the frequency distribution table which shows the marks of 100 students in Mathematics examination

Marks	40-50	50-60	60-70	70-80	80-90	90-100
Number of students	10	22	34	25	7	2

Calculate

(i) The mean given that the assumed is 75

(ii) The median in two decimal places.

12.(a) Sketch a cuboid ABCDPQRS such that AB=12cm , AB=8cm and AP=6cm

(i) Find the angle between AR and the plane ABCD to the nearest degree in 12(a) (i) above .

(b) P and Q are two points on latitude 60°N. The distance between P and Q measured along the parallel of latitude is 1500km. Find the difference between their longitude . (Use Π=3.14, R=6370km).

13.(a) For what values of x in which matrix B has no inverse.

$$B=\begin{pmatrix} x-2 & x-2 \\ & \end{pmatrix}$$

3 x

(b) Solve the following pair of simultaneous equations by matrix method ;

$$\begin{cases} 5x - 3y = 7 \\ 2x - 1 = y \end{cases}$$

(c) A matrix T maps the point (2,3) in to (8,18) and the point (-3, -2) in to (-7, -17). Find the Matrix T.

14.(a) The function f is defined by:

$$F(x) = \begin{cases} -2 \text{ if } x < 0 \\ 4 \text{ if } x = 0 \\ X + 1 \text{ if } x > 0 \end{cases}$$

Sketch the graph of f and state the domain and range of f

(b) A wholesaler sells small tins of tomato paste at sh. 960 per dozen and those of chilli sauce at sh. 768 per dozen. A customer decides to buy at least one dozen of each, but she cannot carry more than 6 dozens altogether and must not spend more than sh. 4800 on the shopping . If the seller makes a profit of sh 384 per dozen tins of tomato paste and sh. 192 per dozen tins of chilli sauce , how many dozens of each commodity must he sell to maximize his profit?

THE END